

We consider the unconstrained optimization problem

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x), \quad (1)$$

where $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuously differentiable nonconvex function, and $x \in \mathbb{R}^n$ is the optimization variable. We focus on settings where objective function evaluations are contaminated by bounded and non-diminishing numerical noise [12]. Specifically, only inexact function values $\tilde{f}(x)$ are available for optimization. Such inaccuracies are unavoidable in many realistic scenarios, including finite-precision arithmetic, simulation-based evaluations, and stochastic approximations [1, 7, 11]. When function values are contaminated by numerical noise, standard optimization algorithms may lose their theoretical guarantees, behave erratically, or terminate prematurely with unreliable solutions.

In the absence of noise and with accurate function evaluations, quasi-Newton methods, in particular the Broyden–Fletcher–Goldfarb–Shanno (BFGS) method and its Limited-memory variant (L-BFGS), are among the most efficient and widely used algorithms for solving problem (1) [6, 8]. Their success is largely attributed to fast local convergence and scalability. Standard implementations [9, 15] rely on line search procedures that enforce the Wolfe conditions to guarantee sufficient descent and the positive definiteness of the Hessian approximation.

In the presence of errors in function values, the sufficient-decrease test used in these line searches may become unreliable. Differences in function values can be dominated by numerical errors, leading to unstable step sizes, ill-conditioned Hessian approximations, or abnormal termination [10, 11]. From a computational perspective, improving the stability of quasi-Newton methods in noisy environments is therefore crucial. In particular, it is desirable to design algorithms that (i) retain computational efficiency when function evaluations are accurate, (ii) remain stable when function values are unreliable, and (iii) guarantee convergence to first-order stationary points.

In this work, we propose a noise-tolerant regularized quasi-Newton method that addresses these challenges. Our approach combines an Armijo line search, quadratic regularization [5, 14], and an Objective-Function-Free Optimization (OFFO) inspired strategy [3, 4]. When a sufficient decrease in function value is observed, the method behaves similarly to standard quasi-Newton methods with line search, preserving its practical efficiency. When numerical noise dominates, we introduce regularization on the approximate Hessian to ensure numerical stability, which yields convergence properties similar to those of OFFO methods, particularly AdaGrad-Norm [16]. The resulting algorithm is particularly well-suited for low-precision or noisy environments. Extensive numerical experiments on the CUTEst benchmark collection, across various floating-point precisions and artificial noise settings, show stable performance under function-value noise and reduced precision.

The key methodological advance is to switch between unregularized quasi-Newton steps and regularized steps using an OFFO-based cumulative-gradient scale. Using this scale at every iteration avoids reliance on measured function decreases when controlling regularization but can be unnecessarily conservative when function values are reliable. Conversely, merely adding a nonvanishing error allowance to a line search does not guarantee convergence to arbitrarily small first-order tolerances be-

Method	Uses f	Noisy f	Noisy ∇f	Nonconvex	$\mathcal{O}(\varepsilon^{-2})$ guarantee
L-BFGS [6, 9, 15]	✓	–	–	✓	–
Regularized Newton method [13]	✓	–	–	✓	✓
Regularized L-BFGS method [5]	✓	–	–	✓	–
Noise-tolerant quasi-Newton method [11, 17]	✓	✓	✓	–	–
OFFO method [4]	–	✓	–	✓	✓
Proposed method ??	✓	✓	–*	✓	✓

Table 1: Theoretical comparison of the proposed method and related Newton-type, quasi-Newton, and OFFO methods. A checkmark (✓) denotes theoretical coverage and “–” its absence. The “Noisy f ,” “Noisy ∇f ,” and “Nonconvex” columns indicate whether the corresponding convergence theory allows function-value errors, allows gradient errors, and covers nonconvex objectives, respectively. The asterisk indicates that gradient noise is considered in numerical stress tests, but its analysis is outside the scope of this paper.

cause its accumulated contribution need not be bounded. Our test permits an unregularized step only when error-adjusted function values certify progress and otherwise activates the OFFO-based scale. The test bounds the accumulated allowances over unregularized iterations, while the cumulative-gradient scale controls the remaining iterations without requiring measured function decreases. This combination retains unregularized quasi-Newton steps and establishes an $\mathcal{O}(1/\varepsilon^2)$ worst-case iteration complexity under persistent function-value errors, matching the optimal worst-case order for smooth nonconvex problems with Lipschitz-continuous gradients [2].

0.1 Comparison With Related Work

We next compare our method with related quasi-Newton and OFFO methods, whose theoretical guarantees are summarized in Table 1.

The closest methods are regularized L-BFGS method [5] and Noise Tolerant Quasi-Newton (NTQN) [11]. The regularized L-BFGS method assumes reliable function values and establish global convergence to first-order stationarity [5, Theorem 2]. Shi et al. [11] develop NTQN for bounded errors in both function and gradient evaluations, using a noise-tolerant line search and a lengthening procedure to stabilize the quasi-Newton updates. Their analysis allows bounded gradient errors and establishes convergence to a noise-dependent neighborhood for strongly convex objectives [11, Theorem 3.5]. Our relaxed Armijo test is related to their noise-tolerant test [11, Section 4], whereas our switch controls the accumulated error allowances and yields an $\mathcal{O}(1/\varepsilon^2)$ bound for arbitrarily small first-order tolerances on smooth nonconvex objectives with accurate gradients.

Regularized Newton and OFFO methods provide further theoretical context [3, 4, 13]. Regularized Newton methods attain an $\mathcal{O}(1/\varepsilon^2)$ iteration bound with reliable function values [13, Theorem 4.1], while OFFO methods avoid function values and attain the same order for smooth nonconvex objectives [3, 4]. Our method instead uses inexact function values to permit unregularized quasi-Newton steps when progress is certified and retains this worst-case order under persistent function-value errors.

0.2 Organization of the Paper

References

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